

Day 1 – Python Fundamentals (Complete Lecture)

1. Introduction to Python

Python is a high-level, easy-to-learn programming language used in web development, automation, AI, and data science.

2. Output in Python (print)

The `print()` function is used to display output. Example: `print('Hello World')` Special characters: `\n` (new line), `\t` (tab space).

3. Input in Python

The `input()` function takes user input as a string. We convert it using `int()` or `float()`.

4. Variables

Variables store data. Example: `num1 = 10`, `name = 'Rahul'`.

5. Data Types

`int (10)`, `float (10.5)`, `str ('Hello')`.

6. Operators

Arithmetic operators: `+`, `-`, `*`, `/`, `%`. Used in calculator and even/odd programs.

7. Conditional Statements

`if`, `elif`, `else` are used for decision making. Example: `if num > 0: print('Positive')`

8. Logical Operators

`and`, `or`, `not` are used in conditions. Example: `if username == u and password == p`

9. Error Handling

`try-except` prevents crashes. Example: `try: int(input()) except ValueError: print('Invalid input')`

10. f-Strings

Used for formatted output. Example: `print(f'Result: {value:.2f}')`

11. Programs Covered

Simple Interest Calculator Login System Positive/Negative Checker Even/Odd Checker Largest of Two Numbers Basic Calculator Temperature Converter Swap Two Numbers

12. Key Learnings

You learned input/output, conditions, operators, error handling, and real-world logic building.

13. Conclusion

Day 1 builds the foundation of programming. Understanding these concepts clearly will help in advanced topics.